

Loghman Samani

✉ samani10qman91@gmail.com | 🌐 loqmansamani.github.io

🌐 loghman-samani | 🔄 LoqmanSamani | 📧 @loqmansamani | ✖ Loqman_Samani

📍 Ludwigsburg, Baden-Württemberg, Germany

RESEARCH INTERESTS

My research focuses on computational structural biology and deep learning, with an emphasis on generative modeling for protein structure design, prediction, and interpretation. I am particularly interested in developing physics- and geometry-informed generative models, such as diffusion-based approaches, to advance our understanding of protein folding and dynamics. In parallel, I explore connections with cognitive neuroscience, aiming to leverage insights from both artificial and biological neural systems to develop more interpretable and biologically grounded AI models.

EDUCATION

- **University of Stuttgart** Apr 2023 - Jun 2025
Master of Science in Technical Biology
◦ GPA: 1.6 Stuttgart, Germany
- **University of Kurdistan** Sep 2014 - Jun 2018
Bachelor of Science in Cell and Molecular Biology
◦ GPA: 15/20 Sanandaj, Iran

EXPERIENCE

- **Institute of Cell Biology and Immunology (IZI)** 🌐 Jun 2024 - Jul 2024
Cell Biology Intern Stuttgart, Germany
 - Utilized CRISPR/Cas9 for gene function analysis and studied cancer pathways
 - Conducted flow cytometry to analyze cell death modalities and cell viability (MTT assays)
 - Performed image processing and analysis using FIJI software and time-lapse imaging
 - Operated confocal microscopy to capture high-resolution cellular images

PROJECTS

- **CppNet: A High-Performance C++ Deep Learning Library** Jan 2025 - Present
Tools: C++, Eigen, OpenMP, CUDA [🌐][🔗]
 - Developed a modular deep learning library with layers, losses, optimizers, and metrics.
 - Optimized performance using Eigen for tensor ops and OpenMP for parallelism.
 - Implemented CUDA acceleration for large-scale model training.
 - Designed extensible APIs for custom components and training utilities.
- **TorchDiff: A PyTorch Library for Diffusion Models** May 2025 - Present
Tools: Python, PyTorch [🌐][🔗]
 - Developed a PyTorch library for diffusion models (DDPM, DDIM, SDE, LDM, unCLIP).
 - Enabled conditional and unconditional generation for text-to-image synthesis.
 - Designed modular components for easy use and extension.
 - Integrated utilities like Noise Predictor, Text Encoder, and evaluation metrics.
- **GRN-Designer: Automatically Design and Optimize Gene Regulatory Networks** Apr 2024 - Feb 2025
Tools: Python, PyTorch, Evolutionary Algorithms, PDEs [🌐][🔗]
 - Built an AI-driven framework using evolutionary and gradient-based optimization.
 - Simulated spatial gene expression with PDE-based models.
 - Optimized regulatory parameters for accurate pattern formation.
 - Applied hybrid optimization to improve efficiency and adaptability.
- **Deep Learning: Models, Articles, and Implementations** Jan 2024 - May 2025
Tools: Python, PyTorch, TensorFlow [🔗]
 - Built 15+ deep learning models, including CNNs, RNNs, and Transformers.
 - Developed a GPT-2-like model from scratch to explore LLM architectures.
 - Implemented optimization techniques like SGD, Adam, and Learning Rate Decay.
 - Published two articles on Towards Data Science explaining deep learning concepts.

• **Biostoch: Stochastic and Deterministic Biological Simulations**

Apr 2023 - Jun 2024

Tools: Python, NumPy, SciPy



- Developed a Python library for simulating biological systems.
- Implemented SSA, Tau-Leaping, and CLE for stochastic modeling.
- Integrated Euler and Runge-Kutta methods for deterministic simulations.

• **SmartSolve: Simplified Machine Learning Workflows**

Apr 2023 - Jun 2024

Tools: Python, Scikit-Learn, NumPy



- Developed a Python library for streamlined ML model training and evaluation.
- Implemented regression, classification, clustering, and ensemble methods.
- Integrated preprocessing tools for feature scaling and encoding.

SKILLS

- **Programming Languages:** C++, Python, R, SQL
- **Data Science & Machine Learning:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas
- **Bioinformatics & Computational Biology:** Bioconductor, Biopython
- **Numerical Simulation:** Numerical Simulation Techniques
- **Laboratory Techniques:** Cell viability assays (MTT), Confocal Microscopy, Time-lapse imaging, Flow Cytometry, Western Blotting, EVOS epi-fluorescence

HONORS AND AWARDS

- **11th Place, Cell and Molecular Biology Olympiad** 2018
Iran's National Scientific Competition
 - Ranked 11th in the final stage of the Cell and Molecular Biology Olympiad.
 - Demonstrated strong expertise in cell and molecular biology, out of a competitive field.
- **Member of the Elite Students Foundation** 2018
University of Kurdistan
 - Inducted based on ranking 14th in the national Master's exam in Biophysics and 11th in the Olympiad.
 - Gained recognition as one of the top students in the country in the field of Biophysics and Cell and Molecular Biology.

CERTIFICATIONS

- **Deep Learning Specialization** Aug 2024
- **Machine Learning Specialization** Feb 2024
- **Mathematics for Machine Learning and Data Science Specialization** Jan 2024
- **Genomic Data Science Specialization** Nov 2023

ADDITIONAL INFORMATION

Languages: Kurdish (Native), Persian (Fluent), English (Fluent), German (Fluent)
Interests: Mathematics, Quantum Physics, Cognitive Neuroscience, Martial Arts, Chess, Movies

REFERENCES

- Reference Person 1**
Job Title, Department
Organization/Institution Name
Email: email1@example.com
Phone: +X-XXX-XXX-XXXX
Relationship: [e.g., Thesis Advisor, Manager, etc.]
- Reference Person 2**
Job Title, Department
Organization/Institution Name
Email: email2@example.com
Phone: +X-XXX-XXX-XXXX
Relationship: [e.g., Project Supervisor, Colleague, etc.]
- Reference Person 3**
Job Title, Department
Organization/Institution Name
Email: email3@example.com
Phone: +X-XXX-XXX-XXXX
Relationship: [e.g., Mentor, Collaborator, etc.]